

Soichiro Nishimori

✉ nishimori@ms.k.u-tokyo.ac.jp

🌐 <https://nissymori.github.io/>

Research Summary

RL enthusiast, who is interested in scaling agents across computation, interaction, and data. Toward this goal, I design high-throughput JAX-based environments, exploration-oriented policy objectives, and robust algorithms for imperfect supervision, including noisy preferences and uncurated data.

Education

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| Apr. 2024 – Present | 📖 Doctoral Program in Engineering, Graduate School of Frontier Sciences, The University of Tokyo, Chiba, Japan.
Supervisor: Professor Masashi Sugiyama.
Theme: <i>Reinforcement Learning under Weak Supervision.</i> |
| Oct. 2023 – Feb. 2026 | 📖 Visiting Ph.D. Student, School of Computer Science, The University of Sydney, Sydney, Australia.
Supervisor: Professor Tongliang Liu. |
| Apr. 2022 – Mar. 2024 | 📖 Master of Engineering, Graduate School of Frontier Sciences, The University of Tokyo, Chiba, Japan.
Supervisor: Professor Masashi Sugiyama.
Theme: <i>Offline Reinforcement Learning under Weak Supervision.</i> |
| Apr. 2018 – Mar. 2022 | 📖 Bachelor of Integrated Human Studies, Kyoto University, Kyoto, Japan.
Supervisor: Professor Shin Ishii.
Theme: <i>Partially Observable Markov Decision Process (POMDP).</i> |

Research Publications

International Conference

- 1 **S. Nishimori** et al., “Emergence of exploration in policy gradient reinforcement learning via retrying,” in *ICML*, 2026. 📄 URL: <https://arxiv.org/abs/2606.00151>,
Proposed **ReMax**, a policy optimization objective that encourages exploration in policy-gradient RL by optimizing the best return over multiple trials.
- 2 **S. Nishimori**, X.-Q. Cai, J. Ackermann, and M. Sugiyama, “Offline reinforcement learning with domain-unlabeled data,” in *RLC*, 2025. 📄 URL: <https://arxiv.org/abs/2404.07465>.
- 3 Y. Tang, Y. Zhang, J. Ackermann, Y.-J. Zhang, **S. Nishimori**, and M. Sugiyama, “Recursive reward aggregation,” in *RLC*, 2025. 📄 URL: <https://arxiv.org/abs/2507.08537>.
- 4 S. Koyamada, **S. Nishimori**, and S. Ishii, “A batch sequential halving algorithm without performance degradation,” in *RLC*, 2024. 📄 URL: <https://arxiv.org/abs/2406.00424>.
- 5 S. Koyamada et al., “PGX: Hardware-accelerated parallel game simulators for reinforcement learning,” in *NeurIPS*, 2023. 📄 URL: <https://arxiv.org/abs/2303.17503>,
Contributed to the Backgammon environment in PGX, a JAX-based framework for hardware-accelerated board-game simulation.
- 6 S. Koyamada, K. Habara, N. Goto, S. Okano, **S. Nishimori**, and S. Ishii, “MJX: A framework for Mahjong AI research,” in *2022 IEEE Conference on Games (CoG)*, 2022. 📄 URL: https://ieeegames.org/2022/assets/papers/paper_162.pdf.

Journal

- 1 **S. Nishimori**, Y.-J. Zhang, T. Lodkaew, and M. Sugiyama, “On symmetric losses for policy optimization with noisy preferences,” *Transactions on Machine Learning Research (TMLR)*, 2026.  URL: <https://openreview.net/forum?id=cBWGLmSeao>.

Projects

- 1 **S. Nishimori**, S. Okano, K. Habara, S. Koyamada, E. Yu, and M. Sugiyama, *Mahjax: A GPU-accelerated mahjong simulator for reinforcement learning in JAX*. 2025.  URL: <https://github.com/nissymori/mahjax>,
JAX-based Riichi Mahjong environment for pure self-play RL, reaching over **1M steps/second**.
- 2 **S. Nishimori**, *JAX-CORL: Clean single-file implementations of offline reinforcement learning algorithms in JAX*. 2024.  URL: <https://github.com/nissymori/JAX-CORL>.

Preprints

- 1 **S. Nishimori**, S. Okano, K. Habara, S. Koyamada, E. Yu, and M. Sugiyama, “Mahjax: A GPU-accelerated mahjong simulator for reinforcement learning in JAX,” *arXiv preprint arXiv:2605.20577*, 2026.  URL: <https://arxiv.org/abs/2605.20577>.
- 2 **S. Nishimori** and P. Parmas, “Retry policy gradients in continuous action spaces,” *arXiv preprint arXiv:2606.05888*, 2026.  URL: <https://arxiv.org/abs/2606.05888>.
- 3 S. Ono, J. Ackermann, **S. Nishimori**, T. Ishida, and M. Sugiyama, “Mitigating reward hacking in RLHF via advantage sign robustness,” *arXiv preprint arXiv:2604.02986*, 2026.  URL: <https://arxiv.org/abs/2604.02986>.
- 4 P. Parmas et al., “Ordergrad: Optimizing beyond the mean with order-statistic policy gradient estimation,” *arXiv preprint arXiv:2606.06096*, 2026.  URL: <https://arxiv.org/abs/2606.06096>.
- 5 S. Takashiro* et al., “On advantage estimates for max@k policy gradients,” *arXiv preprint arXiv:2606.06080*, 2026, *Equal contribution.  URL: <https://arxiv.org/abs/2606.06080>.
- 6 B. Tong, J. Komiyama, **S. Nishimori**, and P. Parmas, “Finite-time regret analysis of retry-aware bandits,” *arXiv preprint arXiv:2605.20854*, 2026.  URL: <https://arxiv.org/abs/2605.20854>.
- 7 T. Kitamura et al., “A policy gradient primal-dual algorithm for constrained MDPs with uniform PAC guarantees,” *arXiv preprint arXiv:2401.17780*, 2024.  URL: <https://arxiv.org/abs/2401.17780>.

Experience

Jul. 2022 – Sep. 2023	 Research Intern OMRON SINICX – Tokyo, Japan
Apr. 2024 – Present	 Part-time Researcher RIKEN AIP – Tokyo, Japan
Aug. 2024 – Oct. 2024	 Research Intern Matsuo-Iwasawa Lab, The University of Tokyo – Tokyo, Japan

Fellowships and Grants

Apr. 2024 – Mar. 2027	 Research Fellowship for Young Scientists (DC1) Japan Society for the Promotion of Science (JSPS) JPY 200,000/month; JPY 2,200,000 over three years as research funding
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Fellowships and Grants (continued)

Aug. 2025



Travel Grant

Reinforcement Learning Conference (RLC) 2025

USD 825

Skills

Languages



Japanese, English (TOEFL iBT 96).

Programming



Python, **JAX**, PyTorch, and $\LaTeX$.